

LAB 2 – GPIO Interfacing (Part 1: Output Devices)

1. How does a Common Anode 7-segment display differ from a Common Cathode display?

A Common Anode 7-segment display has all the anodes of the LEDs connected together and tied to VCC (3.3V or 5V). A segment turns ON when its corresponding pin is driven LOW (0), allowing current to flow from VCC through the segment to ground.

In contrast, a Common Cathode display has all cathodes connected together and tied to ground. A segment turns ON when its corresponding pin is driven HIGH (1), allowing current to flow from the microcontroller into the segment.

Key Difference:

Type	Common Pin	Segment Turns ON When
Common Anode	Connected to VCC	Driven LOW (0)
Common Cathode	Connected to GND	Driven HIGH (1)

Therefore, a Common Anode display operates using **active-LOW logic**, while a Common Cathode display operates using **active-HIGH logic**.

2. What is the advantage of using a 74LS247 over directly interfacing a 7-segment display with a microcontroller?

The 74LS247 BCD-to-7-segment decoder simplifies the interface between the microcontroller and the display by:

- Reducing the number of required microcontroller pins (only 4 BCD inputs instead of 7 segment control lines)
- Automatically decoding BCD values into the correct segment outputs
- Reducing software complexity
- Simplifying driving requirements while external resistors ensure proper current limiting

Without the decoder, the microcontroller would need to control all 7 segments individually and store segment patterns in software. Using the decoder makes the design cleaner, more efficient, and easier to expand.

3. How did you implement the 1-second delay in your counter program?

The 1-second delay was implemented using a software delay loop based on the 16 MHz system clock of the TM4C123GH6PM microcontroller.

A nested for loop was used to generate a time delay. The inner loop runs approximately 3180 iterations to create about a 1 millisecond delay. The outer loop repeats this process n times, where n represents the total delay in milliseconds.

The delay function used in the program is shown below:

```
// Delay function (16 MHz clock)
void delayMs(int n){
    int i, j;
    for(i = 0; i < n; i++)
        for(j = 0; j < 3180; j++)
            {} // 1 ms delay
}
```

For example, calling:

```
delayMs(1000);
```

produces approximately a 1-second delay.

The loop value (3180) was determined experimentally to approximate a 1 ms delay at a 16 MHz clock frequency. Although this method is simple and effective, it is not as accurate as using a hardware timer such as the SysTick timer.

4. What were the most common issues you faced when interfacing the TIVA LaunchPad with the 7-segment display?

Some common issues encountered during the lab included:

- Incorrect wiring between PB0–PB3 and the 74LS247 inputs
- Not sharing a common ground between the TIVA LaunchPad, decoder, and display
- Forgetting to enable the clock for Port B
- Incorrect configuration of GPIO pins as outputs
- Display showing incorrect numbers due to reversed BCD bit order

These issues were resolved by carefully checking wiring connections, verifying the pin configuration in the code, and ensuring the correct BCD mapping was used.

5. How did/could you verify that the correct BCD values were sent to the 74LS247?

The correct BCD values were verified by:

- Comparing the output with the truth table provided in Table 2
- Using the debugger in Keil to monitor the Port B data register
- Checking PB0–PB3 voltage levels using a multimeter
- Confirming that displayed digits matched the expected decimal values

If the displayed number matched the intended digit, it confirmed that the correct BCD value was transmitted to the decoder.

6. How could this knowledge be applied to real-world embedded systems applications?

The concepts learned in this lab are applicable to many real-world embedded systems, such as:

- Digital clocks
- Calculators
- Elevator floor indicators
- Scoreboards
- Industrial measurement displays
- Timer systems
- Embedded control panels

These applications require reliable GPIO configuration, digital interfacing, and proper timing control, which were fundamental concepts practiced in this lab.

7. If you were to expand this project, what additional features or components would you include?

If this project were expanded, the following features could be added:

- Push buttons to manually increment or decrement the counter
- Multiple 7-segment displays to create a multi-digit counter
- Implementation of a hardware timer (SysTick) instead of a software delay
- An LCD display for more advanced output
- A countdown timer feature
- Serial communication to send numbers from a PC

These additions would increase functionality and further develop embedded system design skills.